



THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK EXAMINATION

031

PHYSICS 1

TIME 3:00 HOURS

YEAR: 2026

INSTRUCTIONS

1. This paper consists of sections **A**, **B** and **C** with a total of eleven (11) questions.
2. Answer **ALL** questions in section **A** and **B** and any **two (2)** questions from section **C**.
3. Mathematical tables and non-programmable calculators **may** be used.
4. Cellular phones are **not allowed** in the examination room.
5. Write your **Examination number / Names** on every page of your answer booklet(s).
6. Where necessary the following constants **may be** used:
 - (i) Acceleration due to gravity = 10 m/sec^2
 - (ii) Density of water = 1 g/cm^3 or 1000 Kg/m^3
 - (iii) Velocity of sound in air = 340 m/sec
 - (iv) Atmospheric pressure = $102,000 \text{ N/m}^2$
 - (v) Specific heat capacity of water = 4200 J/KgK

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SECTION A: (16 MARKS)
Answer ALL questions in this section

1. For each of the items (i) – (ii), choose the correct answer among the given alternatives and write its letter besides the item number in the answer booklet provided.
- (i) Which of the following best explains why objects submerged in water appear to be closer to the surface than they actually are?
- A. Light travels faster in water than in air, causing the object to appear closer.
 - B. Light rays from the object undergo total internal reflection before reaching the observer.
 - C. Light rays bend towards the normal as they pass from water to air, making the object appear raised.
 - D. The refractive index of water is lower than that of air, causing the light to diverge.
 - E. Light is scattered more in water, resulting in magnification of the object.
- (ii) Which of the following best describe why the knowledge of physics is necessary to understand all other science?
- A. Physics explains how gravity works
 - B. Physics explains how rocks are formed
 - C. Physics explains fundamental aspects of universe
 - D. Physics explains how energy passes from one object to another
 - E. Physics explains the motion of objects that can be seen with naked eyes
- (iii) On a frosty day the metal handlebars of the bicycle feel colder than the rubber grips because
- A. The rubber is better absorber of radiation than metal
 - B. The metal is colder than the rubber
 - C. The rubber has a higher heat capacitor than metal
 - D. The metal is better conductor of heat than rubber
 - E. The expansion of rubber is higher than the metal handle bars
- (iv). When a metal rod is twisted, the internal force that resists the twisting motion is classified as
- A. Shear force
 - B. Torsional force
 - C. Tensile force
 - D. Compressive force
 - E. Bending force
- (v). A stone dropped from the top of a building hit the ground 4 second later. What was its velocity when hitting the ground?
- A. 400m/s
 - B. 45m/s
 - C. 40m/s
 - D. 4.5m/s
 - E. 0.4m/s
- (vi). Which property of concave mirror is suitable for a dentist to consider when selecting concave mirrors for repairing dental related activities?
- A. The one that produces diminished images
 - B. The one with wider field of view
 - C. The one that produces virtual and erect image
 - D. The one that produces larger magnification
 - E. The one in which objects and images are seen easily
- (vii). A force of 15N acts on an object at an angle of 60° to the direction of motion. The object moves 4m the workdone is
- A. 30J
 - B. 60J
 - C. 120J
 - D. 20J
 - E. 240J

- (viii). Why are magnets often fitted to the doors of refrigerators and some of the cupboards?
- A. To keep away from heat B. To keep the inside environment warm
C. To keep the doors tightly closed D. To keep iron away
E. To keep the inside environment cold

- (ix). Which of the following is (are) true both for light waves;

1. They can travel through a vacuum
2. Speed of wave = frequency $\times$ wavelength
3. They transfer energy from one place to another

A. 1,2,3 B. 1,2 C. 2,3 D. 1 only E. 3 only

- (x). Two blocks of equal mass are placed on a table. Block A has a larger base area than block B. which statement is correct

- A. Both exerts equal pressure B. Block A exerts greater pressure
C. Block B exerts greater pressure D. Pressure depends only on mass
E. Block B exerts lower pressure

2. Match the applications of the electromagnetic spectrum in **list "A"** with the corresponding part of electromagnetic spectrum in **list "B"** by writing the letter of the correct response beside the item number in the answer booklet provided

S/N	LIST A:	LIST B:							
(i.)	Can be used to sterilize fruits and vegetables.	Increase in frequency → <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr></table>	A	B	C	D	E	F	G
A	B		C	D	E	F	G		
(ii.)	Stimulates the production of vitamin D in the human skin								
(iii.)	Stimulates the sensitive cells in the retina								
(iv.)	Emitted by a remote control								
(v.)	Used for rapid cooking in an oven								
(vi.)	Used by doctors and dentists to check bones and teeth.								

3. (a). A concave mirror is used to form an image of a pin object. By using a well labelled diagram show how you can obtain a real and same size image as the pin object **(05 marks)**
- (b). An object 2cm high stands on the principal axis of the convex lens at a distance of 9cm from the lens. If the focal length of the lens is 6cm, what is the position and nature of the image? **(04 marks)**
4. (a). The air pressure at the base of a mountain is 76cm of mercury and at the top is 65cm of mercury. Calculate the height of the mountain. **(04marks)**
- (b). Two boys Q and P are carrying a ladder of weight 800N. If Q holds the ladder 2m from its center of gravity and P holds the ladder 3m from its center of gravity. With the aid of a sketch, calculate the load that each of them supports. **(05marks)**

5. (a) Why while swimming, the swimmer pushes the water backward with his hands? **(03 marks)**
- (b) A wheel and axle with an efficiency of 90% is used to raise a lot of 1000N. The radius of the wheel is 40cm while the radius of the axle is 5cm, find the velocity ratio (V.R) and mechanical advantage (M.A). **(06 marks)**
6. (a) The figure below shows overhead powerlines sugging in summer and taut in winter due to temperature changes.

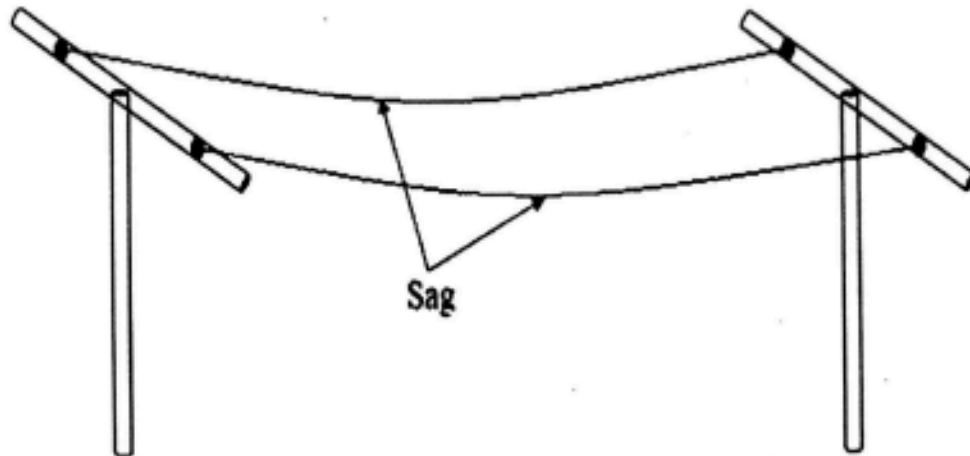


Figure 1

Overhead powerlines are installed with some slack, as shown in figure.

- (i) Explain how thermal expansion and contraction affect the tension in power lines between summer and winter **(03 marks)**
- (ii) What would happen if the powerlines were installed too tightly during summer? **(03 marks)**
- (b) Why particles in a solid expand when left in hot water? Briefly explain by using kinetic theory of matter. **(03 marks)**
7. (a) The fundamental frequency of a vibration of a string is f . What will the fundamental frequency be if the length of the string halved and the tension is increased four times? **(06marks).**
- (b) A radioactive element has a half – life Of 1600 years. what will the fraction of its original mass which has decayed after 6400years be? **(03marks)**
8. (a) Why is the ionosphere considered a crucial layer of Earth's atmosphere for radio communication? **(03 marks)**
- (b) The figure 2 below shows a profile of how altitude(height) varies with temperature
- (i) What does the profile represent? **(01 mark)**
- (ii) Give the name of the layers represented by A,B,C and D **(02 marks)**
- (iii) What is the name given to the zone separating layer A and layer B ? **(01 mark)**

Give one importance for each (02 marks)

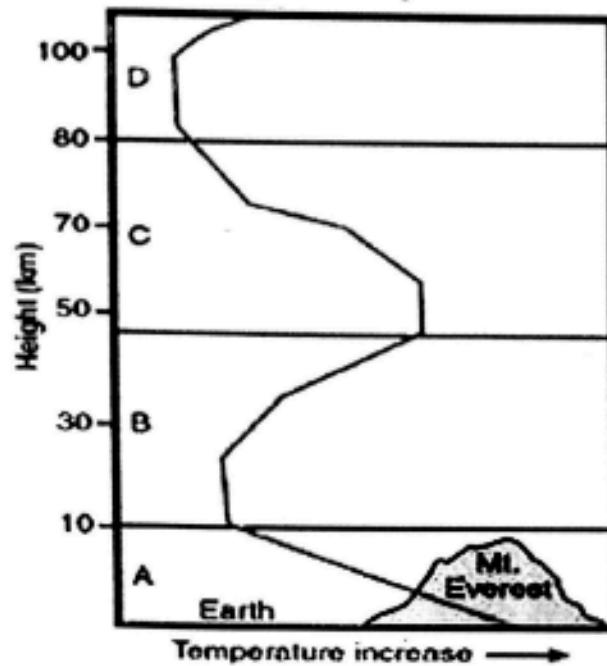


Figure 2

SECTION C: (30 MARKS)

Answer any two (2) questions from this section

9. (a) Consider the circuit in the figure 3 below

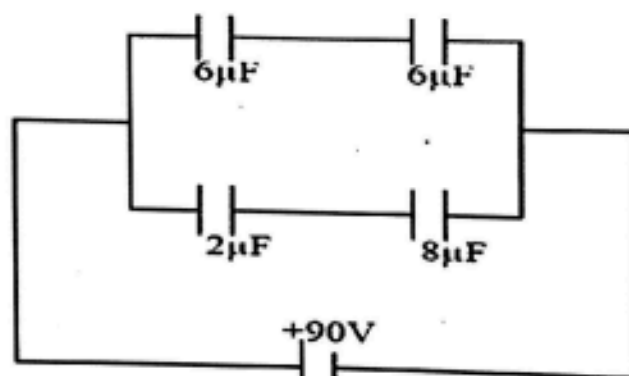


Figure 3

Calculate:

- The equivalent capacitance of the system (03 marks)
 - The total charge flowing through the circuit (03 marks)
- (b). Study the circuit in figure 4 and then answer the questions that follows.

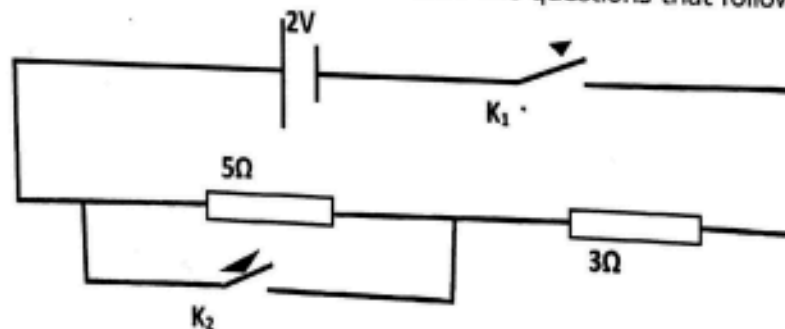


Figure 4

Calculate the current passing through the circuit when:

- (i) Switch k_1 is closed **(03 marks)**
- (ii) Switches k_1 and k_2 are both closed **(03 marks)**
- (iii) Switch k_1 is open and k_2 is closed **(03 marks)**

10. (a) The figure 5 below shows parts of Cathode Ray Oscilloscope (CRO), Study and answer the questions that follows:

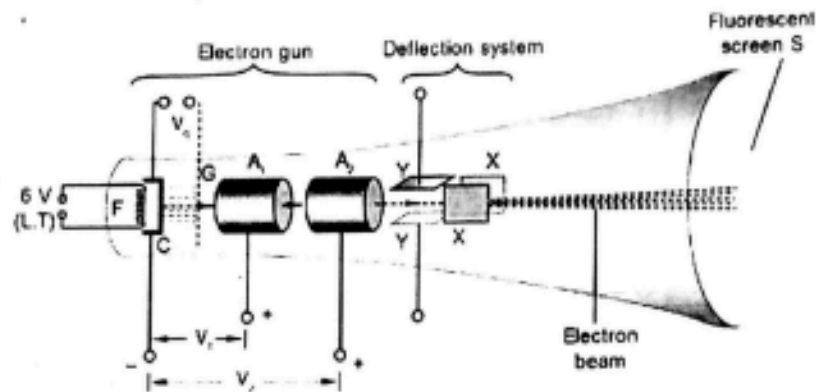


Figure 5

11. (a) (i) How does sound the disco hall transmitted into your room? Briefly explain. **(02 marks)**
- (ii) A man is standing at point P between the walls X and Y shown in Figure 7 and claps his hands once. If the first two echoes are separated by 0.4 s, find the separation distance of the wall X and Y **(05 marks)**
- (b) (a) A student uses a step-down transformer connected to a 240 V a.c. mains supply to operate a low-voltage device. The transformer has 1,200 turns in the primary coil and 200 turns in the secondary coil
- (i) Explain why a transformer works only with alternating current (a.c) and not with direct current (d.c). **(03marks)**
- (ii) Calculate the secondary voltage produced by the transformer. **(03marks)**
- (iii) State two energy losses that occur in a practical transformer. **(02marks)**